



Penitentes on Europa

More about the towering spikes of ice that may exist on the Jovian moon of Europa. <https://digit.in/jul20-61>



Bladed terrain

Research from the New Horizons team exploring the bladed terrain on Pluto, made up of methane ice. <https://digit.in/jul20-62>

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he probes that we have sent out for planetary exploration, as well as the space telescopes

parked in Earth orbit, have all given us glimpses on the surface features of the moons and planets in the Solar System. Even when the imaging resolution of these astronomical instruments are not good enough to directly capture surface images, a range of sensors can give scientists an idea of the surface conditions, which allow them to figure out what kind of features are likely to exist at a given location on another body. For the most part, these worlds are utterly alien, such as the tortured terrain of red striations on the glacial moon of Europa, or the excessive volcanism on Io, or the overlapping craters on Hyperion that looks like it has survived an orbital bombardment. However, at times, the geological features are very similar to the ones on Earth.

There are two ways to look at the geological features described here. Either you can visit these places on Earth and have an experience that is out of this world, exploring the landscapes that are similar to those elsewhere in the Solar System. Or else, any of the lucky locals of these places on the Earth

were to visit the moons and planets in the Solar System, would find themselves in quite familiar surroundings.

NAMIB SAND SEA

Sandwiched between the Eastern coast of Africa and the southern waters of the Atlantic is the Namib sand sea. This is a desert made up of colossal dunes that continuously grow and shift. These are the biggest sand dunes in the world. At a height of 383 meters is Dune 7, the tallest dune in the world, and another popular one known as Big Daddy stands at a height of 325 meters. This is the only desert in the world that is right on the coast. Unlike most sand dunes around the world, which are caused by the erosion of bedrock, the dunes of Namib are created by deposits of winds, rivers and even the sea from the coast. The South African rivers carry the material to the Atlantic Ocean, from where they are deposited back onto the coast. The process has been going on for more than five million years, and continues to this day. Covering an area of three million hectares, the Namib sand sea is a UNESCO world heritage site.

Researchers working with the Curiosity rover have found a sand dune on Mars around Mount Sharp, that is so similar to the ones at the Namib sand

sea, that they named in the Namib dune. It became the first sand dune on another celestial body to be studied. The surrounding landscape was also so similar to the Namib desert, that the location was dubbed as the Windhoek Quadrant, after a similar geological district in Namibia. The scientists studying the sand dunes on Mars have been taking help from the geologists studying the sand dunes in the Namib desert, because of how similar the two formations are. The sand dunes on Mars allow scientists to find out the action of wind on dunes in conditions of low gravity and a thinner atmosphere. Observations from the Mars Reconnaissance Orbiter have showed that the distribution of minerals on the dunes are not uniform, and it is likely that the action of the wind is sorting smaller and larger particles, leading to local accumulations of some kinds of minerals. There is ongoing research on why the ripples on the dunes of Mars are much larger than the ripples on the dunes in the Namib desert. Scientists believe that the difference may have something to do with the lower atmospheric pressure on Mars.

THE LOST CITY

In the middle of the northern Atlantic Ocean is an underground dome-

ALIEN PLANET

An out of this world experience on the Earth

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The Namib sand sea